EMBEDDED SYSTEM WITH IOT
EXPERIMENT NO. 2

Home Appliance Automation using IoT

Name: Lelis, Kathryna T.

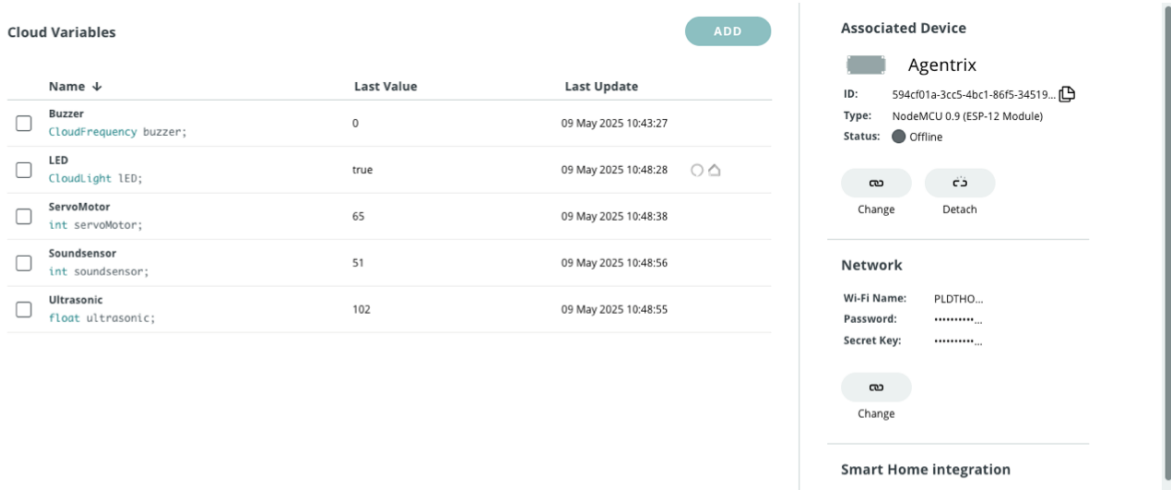
Course/Section: CPE162P-4/E01

Group No.: INDIVIDUAL

Date of Performance: May 5, 2025
Date of Submission: May 9, 2025

Cyrel O. Manlises, Ph.D.
Instructor

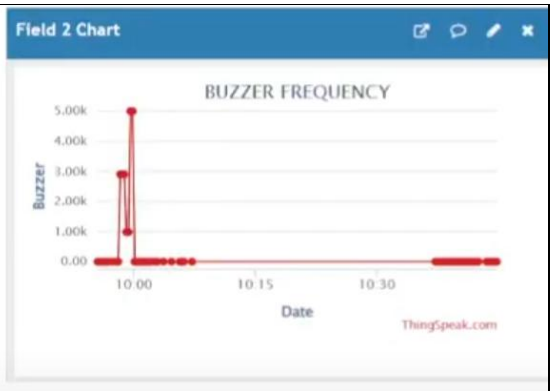
DISCUSSION

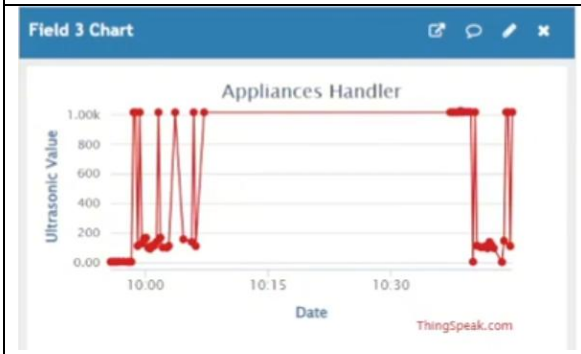


Experiment 2 (Home Appliance Automation using IoT) is an existing experiment from previous term. This time we are tasked to do the same yet with added IoT features including the data gathering such as thingspeak. With this, I found an app that can simplify the way I handle IoT, which is the Arduino IoT cloud. First step in achieving all objectives is to explore this application. In Arduino IoT Cloud I extablish a

dashboard where I can control the said sensor and components I wanted to use. After which, I coded this components in the sketch with overall guidance in a youtube videos.

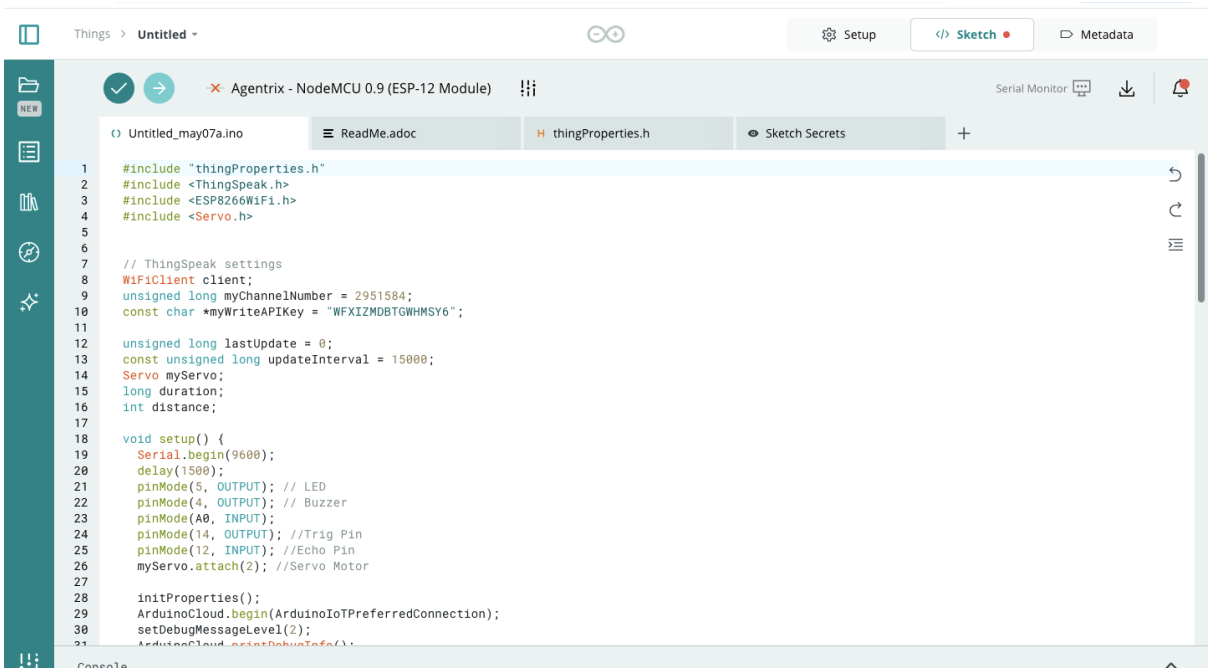
As part of the data analytics requirement, I configured the system to push sensor readings to ThingSpeak. This allowed for real-time logging of the ultrasonic sensor s values, servo motor status, and sound sensor input. Fields were mapped in ThingSpeak to correspond to each component, allowing for meaningful visualization and analysis over time.



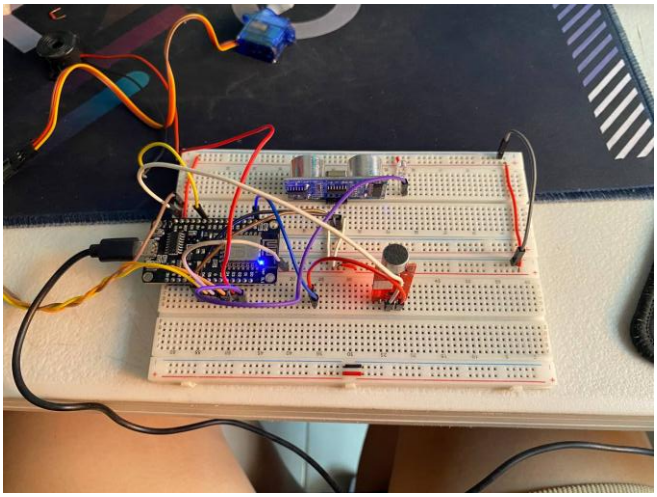


Field 1 shows the LED status toggling between ON and OFF, indicating responsive remote control. Field 2 logs the buzzer's frequency with a sharp spike followed by inactivity, suggesting

brief alerts possibly triggered by sensor input. Field 3s ultrasonic sensor data fluctuates significantly, reflecting real-time distance measurements for object detection. Field 4 captures servo motor rotation, indicating mechanical actuation in response to user commands. Lastly, Field 5 records sound levels with a gradual increase, possibly reflecting environmental noise or user interaction. Together, these graphs demonstrate how sensors and actuators interact with cloud services to enable intelligent automation and data-driven control in a smart home environment.



Once overall coding has been done, the component physically is also being created. Since im using ESP8266 nodemcu and im not familiar with it, I use pinout guidance I researched in google.



CONCLUSION

The experiment was able to successfully fulfill its goals through the practical demonstration of the use of IoT in home appliance automation. It illustrated how wireless communication and embedded systems may be used to automate devices such as sensors, buzzer, LEDs, and servo motors.

Concepts learned from CPE160P and CPE161P e.g., microprocessors, switching theory, and logic circuits were implemented using Arduino coding and hardware configuration, which anchored the project in scholarly bases.

The ESP8266 NodeMCU integration with the Arduino IoT Cloud provided real-time remote control using an online dashboard, while ThingSpeak facilitated effective data collection and analysis of system usage.

The project generally brought theoretical learning into practice, illustrating how IoT technologies can be used to improve common systems such as home automation.